Ashraf Ali Hussain Al-Saloul

Cyberjaya, Malaysia • +60 18-381 6023 • thalththanwyd@gmail.com
LinkedIn: ashraf-ali-hussein • GitHub: ASHRAF-2004 • Portfolio: ashraf-2004.github.io

Summary

Computer Science student specializing in Software Engineering with hands-on experience building backend, systems, and web projects using C++, Python, FastAPI, SQLite, and JavaScript. Targeting Software Engineering or Backend Developer internships involving Application Programming Interface (API) development, database-backed workflows, and applied machine learning prototypes.

Education

Multimedia University, Cyberjaya Campus

Bachelor of Computer Science (Hons.), Specialization in Software Engineering • Expected October 2027 • Cyberjaya, Malaysia

Projects

Machine Learning Based System for Cardiopulmonary Sound Separation • Final Year Project (FYP), In Progress Academic Prototype

Python, FastAPI, PyTorch, SQLite, HTML, CSS, JavaScript • GitHub: Project repository

- Developed a FastAPI academic prototype for cardiopulmonary sound separation, supporting WAV upload, separation requests, result retrieval, output downloads, and recent job history.
- Integrated a PyTorch-based NeoSSNet inference wrapper to process mixed WAV audio, resample inputs to 4000 Hz, and generate separate heart and lung WAV outputs for academic evaluation.
- Designed a SQLite and SQLAlchemy metadata layer with 6 tables for uploaded audio, model configuration, separation jobs, output paths, evaluation metrics, and system logs.

FalconOCR • In Progress

C++17, CMake, Win32 API, Image Processing, GoogleTest • GitHub: Project repository

- Built a from-scratch C++17 Optical Character Recognition (OCR) prototype without external OCR libraries, using CMake, native Win32 GUI support, and a CLI preview for BMP, PGM, and PPM image inputs.
- Implemented OCR pipeline stages for Otsu binarization, connected-component segmentation, glyph normalization, and template classification.
- Added Unicode language-pack discovery and GoogleTest smoke tests for classifier and OCR pipeline behavior.

Unlock PDF • Completed, Planned Enhancement

C++17, CMake, Python/Tkinter • GitHub: Project repository

- Developed a C++17 PDF password recovery utility for user-owned documents with CMake builds, configurable wordlist-based recovery options, password length constraints, and worker threads.
- Implemented PDF metadata parsing and recovery handlers for RC4 and AES-based standard security variants using custom C++ helper modules.
- Added a Python/Tkinter GUI wrapper and sample encrypted PDFs for manual verification of supported recovery workflows.

Technical Skills

Languages: C++, Python, JavaScript, HTML, CSS
Frontend: React, JavaScript, HTML, CSS
Backend: FastAPI, Python, REST APIs, Application Programming Interface (API) Development
Databases: SQLite, MySQL, SQLAlchemy
Tools: Git, GitHub, VS Code, Docker, CMake, LaTeX, Overleaf
Concepts: Object-Oriented Programming (OOP), Software Design, Software Requirements Engineering, Database Design

Relevant Coursework

Software Design, Software Requirements Engineering, Object-Oriented Programming, Database Systems, Computer Networks, Operating Systems

Languages

Arabic: Native • English: Working proficiency